CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Matthew Le 10/27/2026
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

R03 fails Q01 and Q05. Its Machine_ID is empty and its Inspector_ID is empty, while its Status reads READY. Both rules come from W07-S02: Machine_ID must not be blank, and a READY row must name an inspector. So this row reports a finished, ready inspection that cannot be tied to a specific machine or to the person who signed it off. R03 is also why the five rule counts add to 12 while only 11 rows are affected, since two rules flag the same single row.

Next action: what to verify, who owns it, and how to retest

The Maintenance Records Owner should verify R03's Machine_ID against the original inspection record, and the Shift Supervisor should confirm the READY status and name the inspector. Neither value gets entered until confirmed. After a correction is recorded in Raw, I refresh and recheck Q01 and Q05, which should fall to 1 and 2, with the distinct total dropping to 10.

Limit: why you cannot guess a correction or prove accuracy

I cannot supply the missing Machine_ID myself, because once a guessed value sits in the cell it looks exactly like a verified one and nobody reading the file later can tell them apart. These five rules test completeness, uniqueness, consistency and validity, but not accuracy, which the Government Data Quality Framework defines as how closely data matches reality. A row can pass all five and still describe an inspection that did not happen as recorded.

Professional connection: one framework idea and how you used it

The framework's action plan ends by repeating the measurement with the same assessment method so results can be compared over time. I applied that by preparing the data in Power Query instead of cleaning cells by hand. The Trim and UPPERCASE steps are saved in the query, so after a confirmed correction the same preparation reruns on Refresh All and the new counts are directly comparable to these.

Refresh test: before, changed value, after, and restored result

In this workbook I changed R01 Inspection_Minutes on the Raw sheet from 20 to 0 and chose Data > Refresh All. Q04 rose from 3 flagged rows to 4, and the distinct total rose from 11 to 12, because R01 now failed the 1 to 60 range while still failing nothing else. I then restored 20 and refreshed again, and both figures returned to 3 and 11. This confirms the Audit results are recalculated from the saved query rather than typed in by hand.

Export only Report: two landscape pages, page break before row 26. Review both pages.